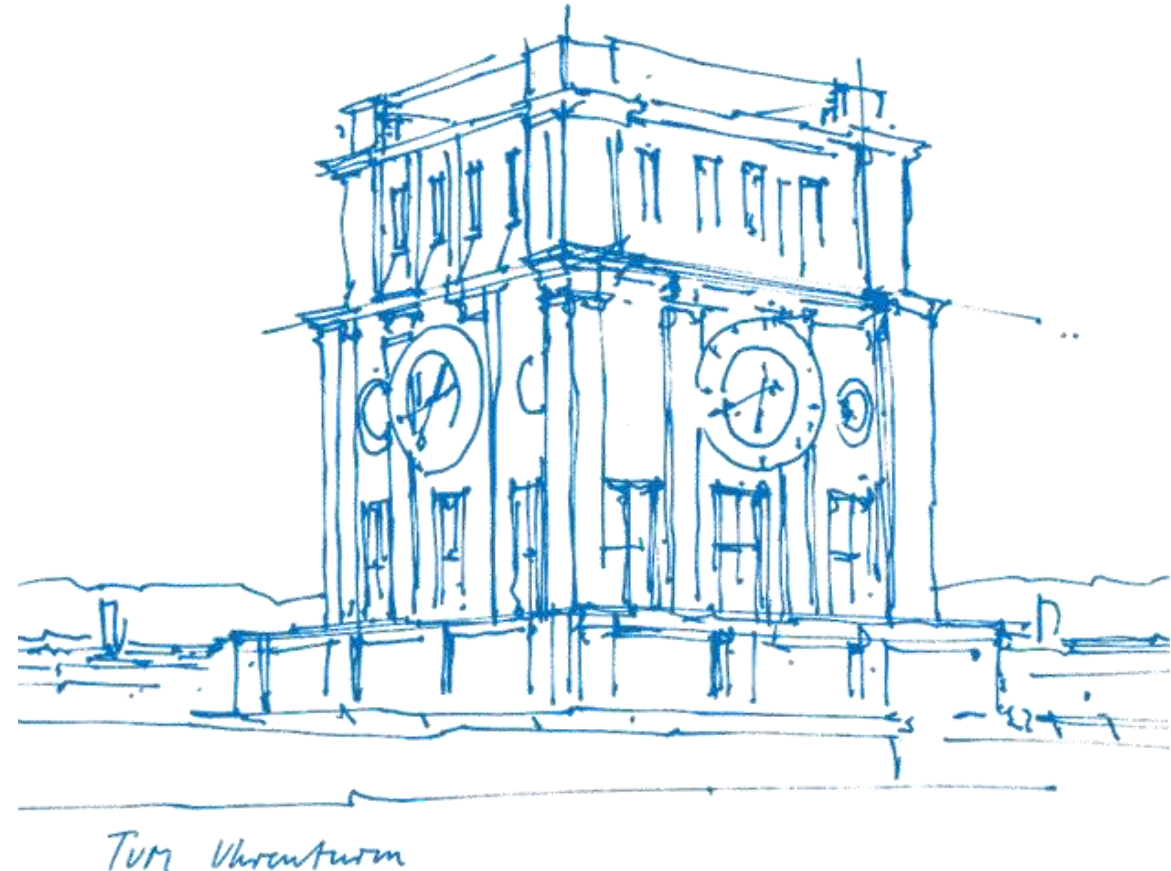


Tutorial 02

Measure Space and Probability Space

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About Tutorial

■ **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
- 14:15-15:45 Tue, 00.08.053 (B42)

■ The slides are not guaranteed to be accurate! (Written by myself)

■ **Materials of this tutorial:** Please scan the QR-Code 

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- **PART I.** Recap of the Lecture
- **PART II.** Exercises from the Exercise Sheet
- **PART III.** Additional Exercises (with solution)



I. Recap

- **Measure and Measure Space**
 - Measure and its properties
 - Measurable space and measure space
- **Probability Measure and Probability Space**
 - Definition
 - Inclusion-exclusion principle (*Siebformel*)

II. Exercise Sheet

Exercise 1. True or false? Justify your answers either by a proof or a counterexample.

- (i) For a probability space $(\Omega, \mathcal{A}, P)$ we have $P(\Omega \setminus A) = 1 - P(A)$ for all $A \in \mathcal{A}$.
- (ii) Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, $A, B \in \mathcal{A}$ such that $\mu(A \cap B) < \infty$. Then $\mu(B \setminus A) = \mu(B) - \mu(A \cap B)$.
- (iii) For a probability space $(\Omega, \mathcal{A}, P)$ and a sequence $(A_i)_{i \in \mathbb{N}}$ in $\mathcal{A}$, we have

$$P\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sup_{i \in \mathbb{N}} P(A_i).$$

Let $S \subseteq \mathbb{R}$. A real number $u \in \mathbb{R}$ is called an *upper bound* of S if $x \leq u$ for all $x \in S$. An upper bound s of S is called *supremum* or *least upper bound* if for all upper bounds u of S in $\mathbb{R}$ we have $s \leq u$. We then write $\sup(S) = s$. When $S = \{x_i \mid i \in \mathbb{N}\} \subseteq \mathbb{R}$ is at most countable, we also write $\sup_{i \in \mathbb{N}} x_i = s$. **Suprema and infima** can be defined for subsets of any partially ordered set.

II. Exercise Sheet

(iv) Let $(\Omega, \mathcal{A}, P)$ be a probability space and $A, B, C \in \mathcal{A}$. Then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

(v) The triple consisting of $\mathbb{N}$, $\mathcal{P}(\mathbb{N})$, and the set function

$$\mu : \mathcal{P}(\mathbb{N}) \rightarrow [0, \infty], A \mapsto \begin{cases} 0 & A \text{ finite} \\ \infty & A \text{ infinite} \end{cases}$$

form a measure space.

(vi) We consider the two-time fair coin flip, where we interpret the two sides to count as 0 or 1. We are only interested in the sum of the two outcomes. Then the cdf is given by

$$F(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{4} & x \in [0, 1), \\ \frac{3}{4} & x \in [1, 2), \\ 1 & x \geq 2. \end{cases}$$

II. Exercise Sheet

Exercise 2. Let $\Omega = \{\square, \begin{smallmatrix} \square \\ \square \end{smallmatrix}, \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}, \begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}, \begin{smallmatrix} \square & \square & \square & \square \\ \square & \square & \square & \square \end{smallmatrix}, \begin{smallmatrix} \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square \end{smallmatrix}\}$, represent the outcomes of rolling of a fair die. Assume a uniform distribution over Ω and let $p \in [0, 1]$ be the probability of obtaining at least one $\begin{smallmatrix} \square \\ \square \end{smallmatrix}$ when throwing six fair dice at once. Compute p (using rigorous mathematical notation, i.e., setting up the probability spaces properly).

II. Exercise Sheet

Exercise 3 (*Inclusion-exclusion formula*). Let $(\Omega, \mathcal{A}, P)$ be a probability space, $(A_i)_{i \in [n]} \in \mathcal{A}^{[n]}$ for a given $n \in \mathbb{N}$, where $[n] := \{1, 2, \dots, n\}$. Then

$$P(A_1 \cup \dots \cup A_n) = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P(A_{i_1} \cap \dots \cap A_{i_k})$$

In particular, it holds that

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2) .$$

[Hint: Use complete induction.]

Bonus: Show that moreover

$$P(A_1 \cap \dots \cap A_n) = \sum_{k=1}^n (-1)^{k-1} \sum_{\{i_1, \dots, i_k\} \subseteq [n]} P(A_{i_1} \cup \dots \cup A_{i_k}) .$$

II. Exercise Sheet



Exercise 4. The DWT tutor Angelika supervises an exercise group with n sheets being submitted anonymously. After grading, Angelika randomly redistributes the sheets. What is the probability that no one gets their original sheet back? How does this probability behave as $n \rightarrow \infty$?

[Hint: Use the inclusion-exclusion formula and that $\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$]

III. Additional Exercises

Exercise 5. After a long party night, Anna and Bob are the only ones left at the Marienplatz subway station, where a train departs every 10 minutes. Unfortunately, the trains are so overcrowded that only one person can board at a time. Each person on the platform has an equal probability of successfully boarding. In addition, two new people arrive at the station during the time before the next train arrives. Determine the probability that Anna

- (a) can board the n -th train,
 - (b) can board no later than the n -th train,
- where $n \in \mathbb{N}$.

(a) Let event A_n be "Anna boarded on the n -th train". Everytime when a train leaves, # people increases 1, which means when n -th train arrives, there are $n+1$ people waiting on the platform. To "board the n -th train", Anna has to miss first $(n-1)$ trains.

$$P(A_1) = \frac{1}{2}. \quad P(A_2) = (1 - \frac{1}{2}) \times \frac{1}{3} = \frac{1}{6}. \quad \dots \quad P(A_n) = (1 - \frac{1}{2}) \times (1 - \frac{1}{3}) \times \dots \times (1 - \frac{1}{n}) \times \frac{1}{n+1} = \frac{1}{2} \times \frac{2}{3} \times \dots \times \frac{n-1}{n} \times \frac{1}{n+1} = \frac{1}{n(n+1)}.$$

(b) Let event B be "Anna boarded no later than the n -th train",

$$P(B) = P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) = \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)} = 1 - \frac{1}{2} + \frac{1}{2} - \dots + \frac{1}{n} - \frac{1}{n+1} = \frac{n}{n+1}.$$

(Directly added because A_i 's
are independent from each other!)

Materials

